

Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper is the addendum (-AddA) to the parent **Paper-15: Quantum-gates perspective on $SO(10)$ gauge anomalies**, written in response to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Canonical TECT positioning (binding 2026-05-26): *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* TOE-level claim SUSPENDED until non-standard rescue independently derived.

I. BACKGROUND AND PARENT-PAPER RECAP

Paper-15 presented the GAP-3 closure (SM anomaly cancellation) within the quantum-gates Stage-2 framework, using Math157's spinor-trace method as the core argument. The previous status was T6 PROVED CONDITIONAL on Pillar 4 sub-task 1 (which provides the $SO(10)$ bundle structure).

II. MATHEMATICAL CONTENT SURVIVAL

The mathematical content of Paper-15 is identical to Math157 (Paper-07-ext). Per Paper-07-ext-AddA, the Math157 result survives as a T7 PROVED group-theoretic lemma conditional on H1 ($SO(10)$ realisation by any mechanism). The Paper-15 perspective adds the Stage-2 framework framing but no new mathematical content.

III. RECLASSIFICATION IMPLICATIONS

Post-reclassification, GAP-3 status: **T4 STRONG EVIDENCE**, conditional on Pillar 6 non-standard-rescue derivation. Same structural status as GAP-2 (Paper-14-AddA). Stage-2 composite tier accordingly remains T3 PROOF SKETCH with two gates (GAP-2, GAP-3) downgraded from T6 to T4 by the Pillar 6 EXHAUSTED cascade.

IV. FORWARD PATH: NON-STANDARD RESCUES

Same as Paper-07-ext-AddA: non-standard Pillar 6 rescue derivation re-enables T6 PROVED CONDITIONAL on the rescue's hypothesis set.

V. CROSS-REFERENCES

- Math157 (canonical anomaly cancellation)
- Paper-07-ext-AddA (Math157 standalone validity)
- Paper-14-AddA (GAP-2 BRST companion addendum)
- Paper-15 (parent paper)
- Math411-AddB §10 (2026-05-26 reclassification)